MERIDIAN: A Longitudinal Clinical Decision Engine for Prescription Safety

Industry / Inhouse Project : 2025-26

Capstone Review - I

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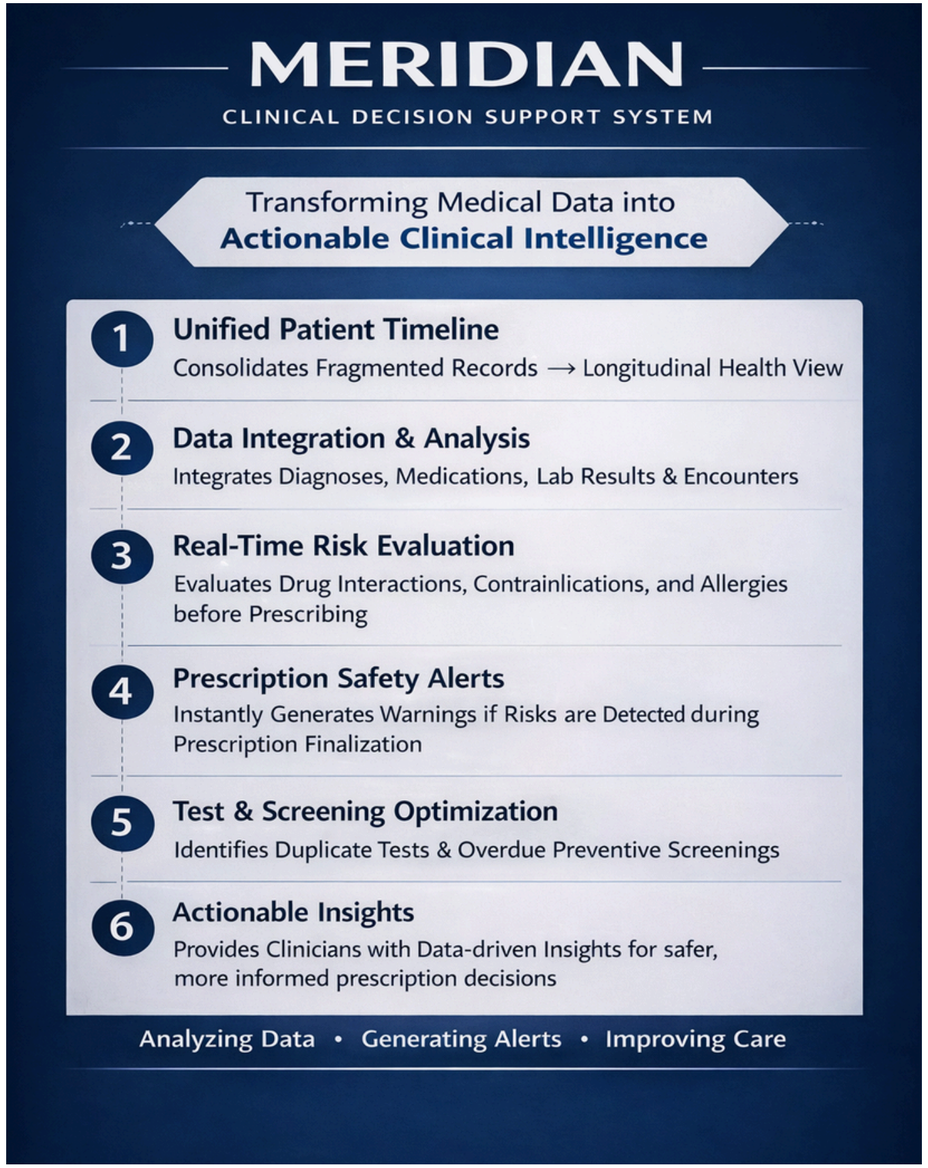
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Introduction to Project

MERIDIAN – Longitudinal Clinical Decision Support System

- MERIDIAN is a clinical decision support system designed to assist doctors in making safer and more informed prescription decisions.
- The system consolidates fragmented patient records into a unified, time-ordered longitudinal health timeline.
- It integrates diagnoses, medications, laboratory results, and clinical encounters into a structured analytical framework.
- Unlike traditional Electronic Health Records (EHRs), MERIDIAN does not only store data it actively analyzes it.
- The system evaluates laboratory trends, medication history, allergies, and disease conditions before a prescription is finalized.
- It simulates prescription risks in real time and generates safety alerts if potential contraindications or interactions are detected.
- MERIDIAN also identifies duplicate diagnostic tests and overdue preventive screenings to reduce unnecessary costs and improve care quality.
- The primary objective is to transform static medical data into actionable clinical intelligence that enhances prescription safety and preventive healthcare.



Lacuna in the existing system

Existing System	Primary Focus	Key Limitation (Gap)
Epocrates	Drug reference & interaction check	No longitudinal patient trend analysis
Medscape / Drugs.com	Drug–drug interaction alerts	Not integrated with full patient history
Lexicomp / Micromedex	Professional interaction screening	Limited lab-trend-based risk simulation
MocDoc CDSS (India)	Guideline-based alerts in HMS	Primarily rule-based, not predictive
Epic / Cerner CDSS	EHR-integrated clinical alerts	Expensive, complex, not widely accessible in India
CarePathways / Enterprise CDS	Protocol & workflow guidance	No prescription simulation before confirmation

Problem Definition

Fragmented Patient Data

- Medical records are spread across different hospitals and labs
- No unified longitudinal patient view
- Doctors lack complete historical visibility

Limited Clinical Analysis

- Consultations focus on current reports only
- Lab trends over time are rarely evaluated
- Medication history is not systematically cross-checked

Prescription Safety Risks

- Drug–drug interactions may be missed
- Drug–allergy conflicts are not automatically flagged
- Lab-based contraindications (e.g., renal risk) may go unnoticed

Inefficient Diagnostic Practices

- Tests are repeated due to lack of centralized data
- Preventive monitoring tests are often overdue
- Increased patient cost and reduced care efficiency

Poor Chronic Disease Monitoring

- Early warning signs are often detected late
- No centralized alert system for high-risk patients
- Preventive care is reactive rather than proactive

Literature Survey

Paper / System	Focus / Limitation	MERIDIAN Advantage
EHR Systems	Stores patient data; lacks trend analysis	Longitudinal timeline with historical insights
CDSS for Prescription Safety	Alerts only for current visit; no full patient context	Risk simulation using entire patient history
Drug Interaction Apps (OpenFDA / DrugBank)	Manual checks; no integrated dashboard	Automated alerts and red-flag summary in dashboard
Hospital Information Systems (HIS)	Paper-heavy, fragmented, limited remote access	Web-based MERN stack, centralized, accessible longitudinal data

Methodology Employed

01 Patient health data is collected and stored in a structured database.

02 Medical events are arranged chronologically to form a longitudinal clinical timeline.

03 Laboratory values are analyzed over time using predefined clinical rules.

04 Clinical alerts and red-flag summaries are generated for decision support.

05 Drug safety and test validity are evaluated to detect risks and gaps.

06 A printable clinical summary is produced for continuity of care.

Hardware, Software, Tools and constraint

Hardware	Software	Tools	Constraints
Laptop/Desktop with minimum 8 GB RAM	Frontend: React.js	Visual Studio Code	Uses simulated patient data only
Internet connectivity	Backend: Node.js, Express.js	Postman	Rule-based clinical logic (no deep learning)
	Database: MongoDB	Git & GitHub	OCR limited to printed lab reports
	Analytical Engine: Python (Flask/FastAPI)	MongoDB Compass	Prototype-level, no live hospital integration
	OCR: Tesseract OCR		
	Visualization: Chart.js / Recharts		

The Block Diagram

Core Architecture Pillars

1. Edge & Orchestration

- Reactive "Thin Client": React.js interface managing real-time state for the Rx Simulation Panel.
- API Gateway: Node.js/Express layer enforcing JWT Auth and request routing.

2. Automated ETL Pipeline

- Asynchronous Ingestion: Parallel streams for CRUD operations and OCR-heavy PDF processing.
- Normalization Layer: Standardizes "noisy" lab data and units for temporal consistency.

3. Longitudinal Persistence

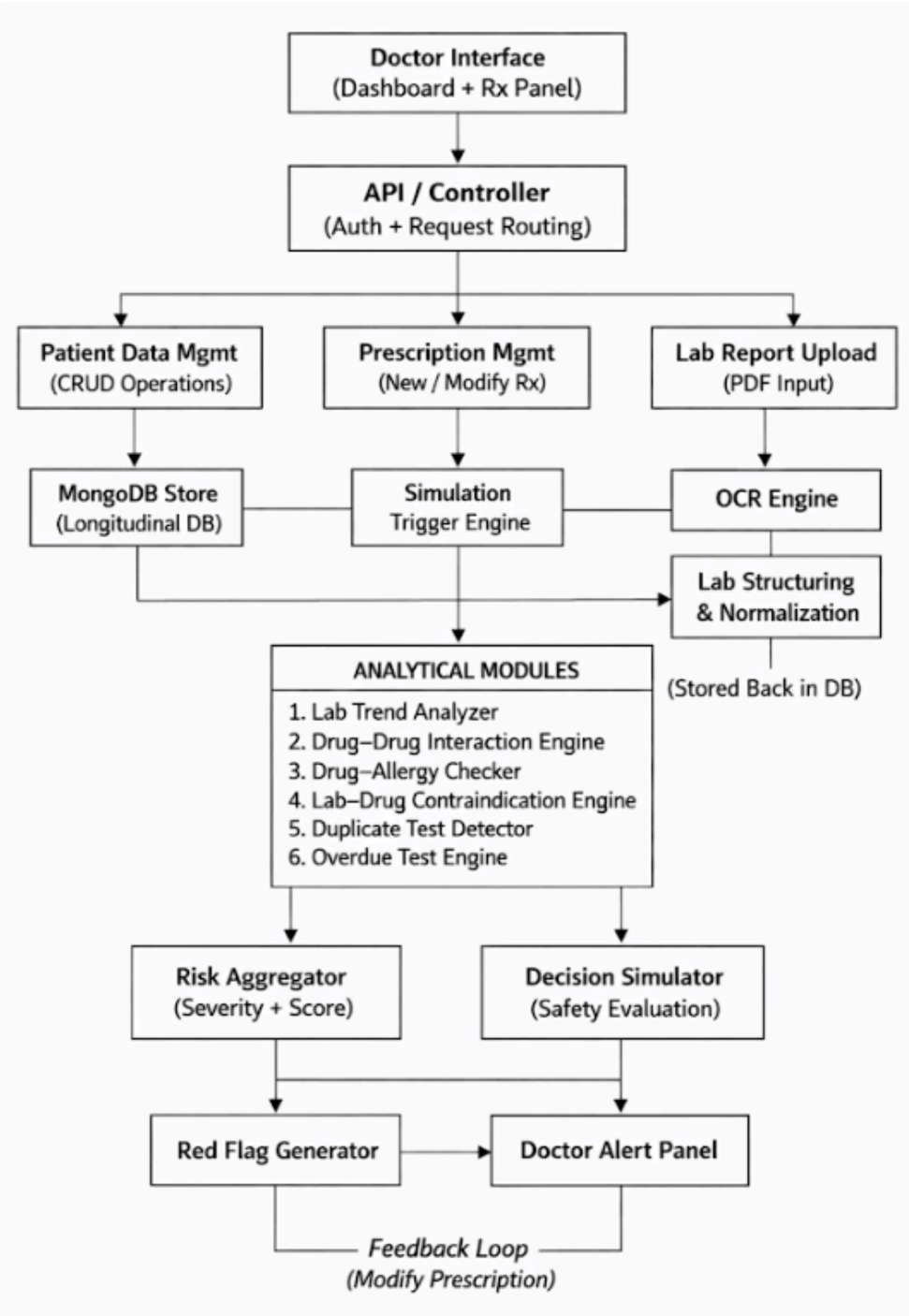
- Time-Series Schema: MongoDB document store optimized for multi-year patient trajectories.
- Context Injection: Queries historical data to feed the engine's trend-analysis logic.

4. Event-Driven Logic Engine

- Pre-Save Simulation: Trigger-based analysis initiated mid-prescription (before DB commit).
- Analytical Suite: Python/FastAPI microservices running parallel Risk Evaluation modules.

5. Heuristic Risk Scoring

- Weighted Aggregator: Converts complex lab/drug data into a prioritized Severity Score.
- Iterative Feedback Loop: Dynamic re-triggering of alerts as the clinician modifies the Rx.



Meridian Workflow

1 Data Ingestion & Longitudinal Storage

- Patient demographics, encounters, and medication history capture
- Laboratory report upload with OCR-based value extraction
- Time-stamped storage in MongoDB (longitudinal schema)
- Mapping with standard reference ranges for abnormality detection

2 Clinical Timeline Builder

- Chronological aggregation of labs, diagnoses, and prescriptions
- Medication start–stop event tracking across visits
- Longitudinal abnormal lab tagging and trend markers
- Generation of a unified patient health profile

3 Backend API & Control Layer

- Role-based authentication for doctor and patient access
- RESTful communication between UI and decision engine
- Alert aggregation and workflow orchestration
- Session-aware prescription evaluation pipeline

4 Analytical Decision Engine

- Multi-visit lab trend analysis using slope and threshold logic
- Drug–drug interaction rule evaluation
- Drug–lab contraindication assessment (e.g., renal function)
- Temporal duplicate test detection
- Condition-driven overdue test identification

5 Prescription Risk Simulator(Core)

- Evaluates new prescriptions against longitudinal patient context
- Integrates medication history, lab trends, and comorbidities
- Risk stratification output: Safe / Caution / Unsafe
- Generates explainable clinical rationale for each flag

6 Alert & Visualization Dashboard

- Consolidated red-flag clinical risk summary
- Interactive trend graphs for key laboratory markers
- Duplicate test cost optimization alerts
- Preventive care reminders for chronic conditions
- Real-time feedback loop after prescription modification

Next Work Plan

Phase 1 – System Design & Data Infrastructure

- Finalize feature scope and module architecture
- Design MongoDB schema for longitudinal records
- Set up MERN project structure
- Implement authentication and patient data APIs

Output:
Functional backend with longitudinal data storage

Phase 2 - Core Intelligence Modules

- Integrate OCR for structured lab value extraction from reports
- Generate time-ordered longitudinal clinical timeline
- Implement lab trend analysis for risk detection
- Develop drug interaction checks with prescription risk alerts

Output:
Clinical decision engine with risk alerts.

Phase 3 – Frontend Integration, Testing & Demo

- Develop doctor dashboard with longitudinal timeline view
- Display risk alerts, red-flag summary, and trend graphs
- Integrate prescription simulation interface
- Perform testing using synthetic patient scenarios

Output:
End-to-end working MERIDIAN prototype ready for demo

Conclusion

The proposed system, ILHR (Intelligent Longitudinal Health Record), presents a structured and decision-aware approach to managing fragmented patient medical data. By integrating longitudinal health record organization with rule-based clinical decision support mechanisms, the system enables lab trend monitoring, drug interaction detection, preventive test reminders, and duplicate test identification. This approach enhances prescription safety, improves chronic disease surveillance, and strengthens continuity of care in outpatient settings. The prototype demonstrates the feasibility of implementing a lightweight, scalable clinical intelligence layer tailored to the needs of evolving digital healthcare infrastructure in India.

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THANK YOU!